

Arithmetic and complexity

Is the matrix multiplication exponent two?

Difficulty: extreme **Importance:** broadly interesting

Topic: arithmetic complexity of dense matrix multiplication

Status: open; admitted after a bounded literature search found no resolution.

Problem statement

Over the field $\mathbb{C}$, let $M(n)$ be the minimum number of scalar additions, subtractions, multiplications, and divisions in an arithmetic straight-line program computing every entry of AB for arbitrary $A, B \in \mathbb{C}^{n \times n}$. Programs must be defined on their intended inputs. Set $\omega = \inf\{\tau : M(n) = O(n^\tau)\}$. Is $\omega = 2$? Equivalently, for every $\varepsilon > 0$, can these products be computed in $O(n^{2+\varepsilon})$ arithmetic operations? This asks about asymptotic arithmetic cost; it does not assert an $O(n^2)$ algorithm or a floating-point stability guarantee.

Why it matters

Matrix multiplication is a basic cost driver for dense NLA.

References and status

M. Bläser, *Fast Matrix Multiplication*, Graduate Surveys 5 (2013), §§1, 5, provides the computational model. E. Dupont et al., *Improving the matrix multiplication exponent with modern optimization and AlphaE-volve* (2026), abstract and §1, reports $\omega < 2.371177$, which does not reach two. Searches for “matrix multiplication exponent 2026” and “omega equals 2 proof” located improvements, not a resolution. **Admitted: no resolution located.**